

RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF FIRST YEAR ENGINEERING
2022-2023-SEM II
ASSIGNMENT I

SUBJECT: BASIC CIVIL AND MECHANICAL ENGINEERING

PART –I (BASIC CIVIL)

1. Classify the building according to NBC.
2. Define plinth and floor area.
3. Define Quarrying and dressing of stone.
4. What is the constituent material of bricks?
5. What do you mean by grade of cement mortar?
6. List the tests conducted on stones.
7. List out different types of bricks.
8. What is the constituent material of cement?
9. List the types of concrete.
10. What are the various forms of reinforcing steel available in market?

PART – B

1. List the properties and uses of good building stone.
2. List the Characteristics of good bricks. Also give its application.
3. What are the different types of cement? Explain their properties and uses.
4. Explain the properties and uses of concrete.
5. What are the different types of steel? Explain their properties and uses.

PART –I (BASIC MECHANICAL)

1. State when an Engine is called IC Engine?
2. What are the main components of IC Engine?
3. Mention the few applications of IC Engine.
4. How do you classify IC engine.
5. What is meant by Carburetion?
6. Explain the term steam generator.
7. Differentiate fire tube Boiler from Water Tube boiler.
8. What is High pressure boiler? Give two examples.
9. Differentiate Boiler Mountings and Boiler Accessories.
10. List the various Boiler Accessories.

PART – B

1. Explain the component parts of an IC engine with a neat sketch.
2. With relevant sketches explain the operation of Four stroke cycle Petrol Engine.
3. Explain the working of Two stroke Diesel Engine with a neat sketch.
4. Explain the constructional features of Cochran boiler with a neat sketch.
5. Explain the working principle of Bobcock and Wilcox boiler with a neat sketch.